

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Physics

2. **Subject Name:** Electromagnetism
Credits: 4

L-T-P: 3-1-0

3. **Pre-requisites:** None

4. **Syllabus and reference books:**

Syllabus:

Vector algebra, Gradient, Curl and Divergence, Electrostatics, Fields and Potentials, Gauss law, Laplace and Poisson equation, Green's Theorem. (16)

Boundary value problems, Method of Images, Multipole expansion, Electric fields in matter, Polarisation. (10)

Magnetostatics, Induction, Magnetic fields in matter, Magnetisation, Scalar and Vector potentials, Maxwell's equations, Gauge transformations, Lorentz gauge, Coulmb gauge. Irrotational and solenoidal currents. Green function for the wave equation. (16)

Reference Books:

1. Electricity and Magnetism : E.M. Purcell
2. Introduction to Electrodynamics: D. J. Griffiths
3. Classical Electrodynamics: J. D. Jackson
4. Classical Theory of Fields: L. D. Landau and E. M. Lifshitz

5. **Lecture-wise break-up:**

Sl. No.	Topic	No. of lectures
1.	Vector algebra, Gradient, Curl and Divergence, Electrostatics, Fields and Potentials, Gauss law, Laplace and Poisson equation, Green's Theorem.	16
2.	Boundary value problems, Method of Images, Multipole expansion, Electric fields in matter, Polarisation.	10
3.	Magnetostatics, Induction, Magnetic fields in matter, Magnetisation, Scalar and Vector potentials, Maxwell's equations, Gauge transformations, Lorentz gauge, Coulmb gauge. Irrotational and solenoidal currents. Green function for the wave equation.	16
Total number of hours		42